# Small Office Network Setup and VLAN Configuration

## 1. Introduction

This project focuses on designing and implementing a small office network using Cisco Packet Tracer. The network is created for a company called Spitertech Solutions. The organization has different departments such as HR, Sales, and IT.

To improve network security and management, VLANs (Virtual Local Area Networks) are implemented. VLANs allow the separation of departments within the same network infrastructure. Inter-VLAN routing is configured using a router to enable communication between different VLANs. A DHCP server is also configured to automatically assign IP addresses to devices.

## 2. Objectives

The main objectives of this project are:

- To design a small office network.
- To implement VLANs to separate departments.
- To configure inter-VLAN routing using a router.
- To enable automatic IP address allocation using DHCP.
- To test network connectivity between different VLANs.

## 3. Tools and Devices Used

**Software used:**

- Cisco Packet Tracer

**Network devices used:**

- Cisco 2911 Router
- Cisco Catalyst 2960 Switch
- Personal Computers (PCs)
- Copper straight-through cables

## 4. Network Design

The network consists of one router, one switch, and three computers representing different departments.

| Department | VLAN ID | Network Address |
|---|---|---|
| HR | VLAN 10 | 192.168.10.0/24 |
| Sales | VLAN 20 | 192.168.20.0/24 |
| IT | VLAN 30 | 192.168.30.0/24 |

Each department is placed in a separate VLAN to reduce broadcast traffic and increase security.

## 5. Implementation

### VLAN Configuration

VLANs are created on the switch and ports are assigned to specific departments.

- VLAN 10 → HR
- VLAN 20 → Sales
- VLAN 30 → IT
- Inter-VLAN Routing

### Switch CLI Commands

```
enable
configure terminal
vlan 10
name HR
vlan 20
name SALES
vlan 30
name IT

interface fa0/1
switchport mode access
switchport access vlan 10
```

```
interface fa0/2
switchport mode access
switchport access vlan 20

interface fa0/3
switchport mode access
switchport access vlan 30

interface g0/1
switchport mode trunk
```

Router-on-a-stick method is used to enable communication between VLANs. Sub interfaces are configured on the router with different IP addresses for each VLAN.

**DHCP Configuration**

DHCP is configured on the router to automatically assign IP addresses to computers in each VLAN network.

Router CLI

```
enable
configure terminal
interface g0/0
no shutdown

interface g0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0

interface g0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0

interface g0/0.30
```

```
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0

ip dhcp pool HR
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1

ip dhcp pool SALES
network 192.168.20.0 255.255.255.0
default-router 192.168.20.1

ip dhcp pool IT
network 192.168.30.0 255.255.255.0
default-router 192.168.30.1
```

**6. Testing and Verification**

The network is tested using the following commands:

- Ping – to check connectivity between devices.
- show vlan brief – to verify VLAN configuration.
- show ip interface brief – to check router interfaces.

Successful ping responses confirm that inter-VLAN communication is working correctly.

**7. Results**

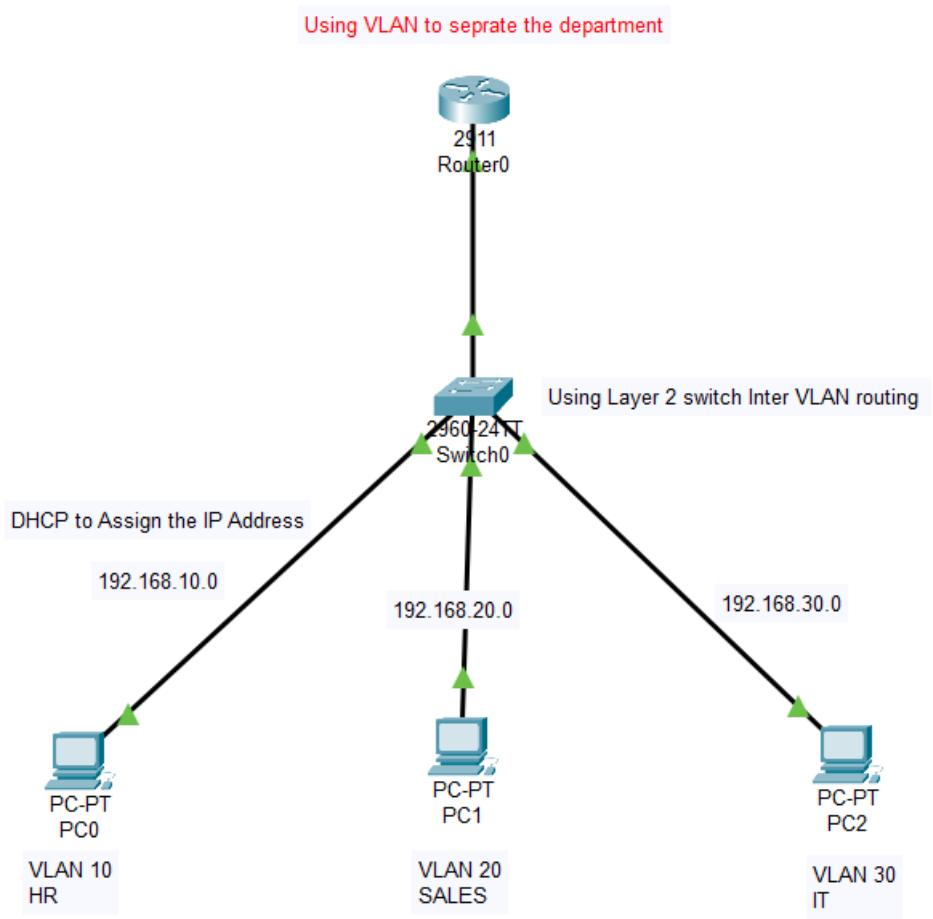
After completing the configuration:

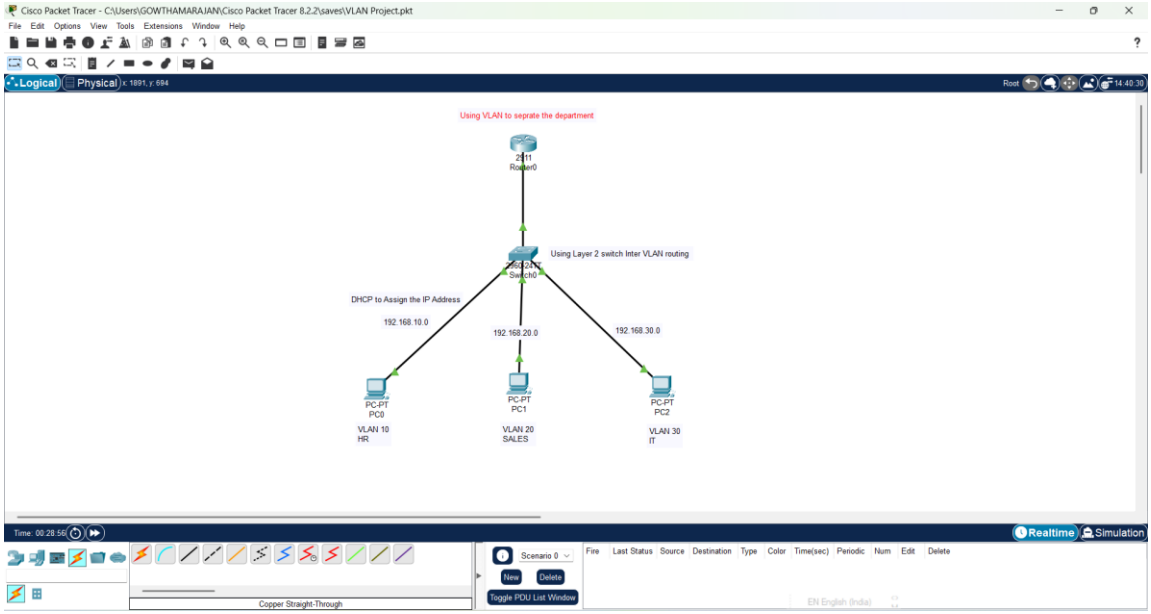
- Departments are successfully separated using VLANs.
- Devices receive IP addresses automatically through DHCP.
- Communication between VLANs is enabled using a router.
- The network operates efficiently and securely.

**8. Conclusion**

This project demonstrates the implementation of VLAN technology in a small office network environment. By separating departments using VLANs and enabling inter-VLAN routing, the network becomes more secure and easier to manage. DHCP configuration simplifies IP address management by automatically assigning IP addresses to devices.

# Screenshots






Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up


Switch>enable
Switch#config te
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name HR
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name SALES
Switch(config-vlan)#vlan 30
Switch(config-vlan)#name IT
Switch(config-vlan)#ex
Switch(config)#interface fa 0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#interface fa 0/2
Switch(config-if)#switchport access vlan 20
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#interface fa 0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#interface g0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

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Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch>show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/2
10   HR                               active    Fa0/1
20   SALES                            active    Fa0/2
30   IT                               active    Fa0/3
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch>
Switch>
Switch>
Switch>show interface trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      1-1005

Port        Vlans allowed and active in management domain
Gig0/1      1,10,20,30

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      1,10,20,30

Switch>
```

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Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no


Press RETURN to get started!



Router>en
Router#config te
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#noshutdown
                  ^
% Invalid input detected at '^' marker.

Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#interface g0/0.10
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#interface g0/0.20
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
```

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Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config-subif)#interface g0/0.20
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#interface g0/0.30
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.30, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up

Router(config-subif)#encapsulation dot1Q 30
Router(config-subif)#ip address 192.168.30.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#ex
Router(config)#ip dhcp pool HR
Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#
Router(dhcp-config)#ip dhcp pool SALES
Router(dhcp-config)#network 192.168.20.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.20.1
Router(dhcp-config)#
Router(dhcp-config)#
Router(dhcp-config)#ip dhcp pool IT
Router(dhcp-config)#network 192.168.30.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.30.1
Router(dhcp-config)#
Router(dhcp-config)#
Router(dhcp-config)#
Router(dhcp-config)#show ip interface brief
                    ^
% Invalid input detected at '^' marker.

Router(dhcp-config)#ex
Router(config)#show ip interface brief
                ^
% Invalid input detected at '^' marker.
```

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Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(dhcp-config)#ex
Router(config)#show ip interface brief
                ^
% Invalid input detected at '^' marker.

Router(config)#ex
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  192.168.10.1    YES manual up                    up
GigabitEthernet0/0.20  192.168.20.1    YES manual up                    up
GigabitEthernet0/0.30  192.168.30.1    YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
Router#
Router#
Router#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
192.168.10.2     00E0.A350.AC65           --                     Automatic
192.168.20.2     000C.85C8.4333           --                     Automatic
192.168.30.2     0005.5E68.55B3           --                     Automatic
Router#









Router con0 is now available
```

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PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.30.2

Pinging 192.168.30.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.30.2: bytes=32 time<1ms TTL=127
Reply from 192.168.30.2: bytes=32 time<1ms TTL=127
Reply from 192.168.30.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.30.2

Pinging 192.168.30.2 with 32 bytes of data:

Reply from 192.168.30.2: bytes=32 time<1ms TTL=127
Reply from 192.168.30.2: bytes=32 time<1ms TTL=127
Reply from 192.168.30.2: bytes=32 time<1ms TTL=127
Reply from 192.168.30.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time=8ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 8ms, Average = 2ms

C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Top